

Control Statements in Java

Aim:

To understand and implement decision-making and looping control statements in Java.

PRE LAB EXERCISE

QUESTIONS

- ✓ List different control statements in Java.

Control statements are used to control program flow.

Types:

Decision: if, if-else, else-if, switch

Looping: for, while, do-while

Jumping: break, continue, return

- ✓ Difference between for, while, and do-while loops.
- **for loop:** used when number of iterations is known.
- **while loop:** checks condition first, may execute **0 times**.
- **do-while loop:** executes first, then checks condition, runs **at least 1 time**.
- ✓ What is the use of break and continue?
 - **break:** stops the loop immediately.
 - **continue:** skips current iteration and goes to next loop.

IN LAB EXERCISE

Objective:

To implement if-else and looping statements.

INPUT STATEMENT:

SCANNER CLASS

- ✓ The Scanner class in Java is used to read input from the user through the keyboard. It is available in the package java.util.
- ✓ The Scanner object reads different types of input such as integer, float, double, and string and stores them in variables.
- ✓ To use the Scanner class, it must be imported before using it in the program.

SYNTAX:

- ✓ `Scanner sc = new Scanner(System.in);`

Commonly Used Scanner Methods:

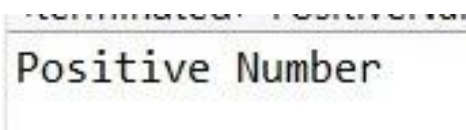
- ✓ `nextInt()` – reads an integer value
- ✓ `nextFloat()` – reads a float value
- ✓ `nextDouble()` – reads a double value
- ✓ `next()` – reads a single word
- ✓ `nextLine()` – reads a complete line of text

PROGRAMS:

Program 1: Check Whether a Number is Positive

```
class PositiveNumber {  
    public static void main(String[] args) {  
        int n = 5;  
        if (n > 0) {  
            System.out.println("Positive Number");  
        }  
    }  
}
```

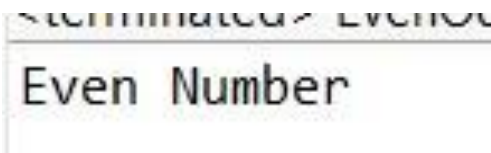
Output:

A screenshot of a Java IDE window titled "Main.java" showing the output of the program. The output is "Positive Number" displayed in a monospaced font within a text area.

Program 2: Check Whether a Number is Even or Odd

```
class EvenOdd {  
    public static void main(String[] args) {  
        int n = 6;  
        if (n % 2 == 0)  
            System.out.println("Even Number");  
        else  
            System.out.println("Odd Number");  
    }  
}
```

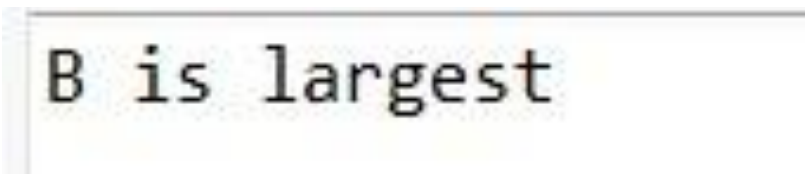
Output:

A screenshot of a Java IDE's output window. The window has a title bar that says "Terminated: EvenOdd". The main content area displays the text "Even Number" in a monospaced font.

Program 3: Find Largest of Two Numbers

```
class LargestTwo {  
    public static void main(String[] args) {  
        int a = 10, b = 20;  
        if (a > b)  
            System.out.println("A is largest");  
        else  
            System.out.println("B is largest");  
    }  
}
```

Output:

A screenshot of a Java IDE's output window. The window has a title bar that says "Terminated: LargestTwo". The main content area displays the text "B is largest" in a monospaced font.

Program 4: Grade Calculation

```
class Grade {  
    public static void main(String[] args) {  
        int marks = 75;  
        if (marks >= 90)  
            System.out.println("Grade A");  
        else if (marks >= 75)  
            System.out.println("Grade B");  
        else if (marks >= 50)  
            System.out.println("Grade C");  
        else  
            System.out.println("Fail");  
    }  
}
```

Output:

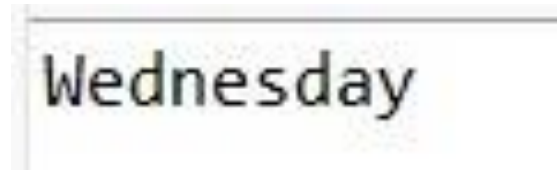
Grade B

Program 5: Day of the Week

```
class DaySwitch {  
    public static void main(String[] args) {  
        int day = 3;  
        switch (day) {  
            case 1: System.out.println("Monday"); break;  
            case 2: System.out.println("Tuesday"); break;  
            case 3: System.out.println("Wednesday"); break;  
            case 4: System.out.println("Thursday"); break;  
            case 5: System.out.println("Friday"); break;  
        }  
    }  
}
```

```
default: System.out.println("Invalid Day");  
}  
}  
}
```

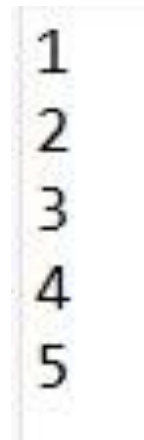
Output:



Program 6: Print Numbers from 1 to 5

```
class ForLoop {  
    public static void main(String[] args) {  
        for (int i = 1; i <= 5; i++) {  
            System.out.println(i);  
        }  
    }  
}
```

Output:



Program 7: Print Numbers from 1 to 5

```
class WhileLoop {  
    public static void main(String[] args) {
```

```
int i = 1;
while (i <= 5) {
    System.out.println(i);
    i++;
}
}
```

Output:

```
1
2
3
4
5
```

Program 8: Print Numbers from 1 to 5

```
class DoWhileLoop {
    public static void main(String[] args) {
        int i = 1;
        do {
            System.out.println(i);
            i++;
        } while (i <= 5);
    }
}
```

Output:

1
2
3
4
5

Program 9: Sum of First 5 Natural Numbers

```
class SumNumbers {  
    public static void main(String[] args) {  
        int sum = 0;  
        for (int i = 1; i <= 5; i++) {  
            sum = sum + i;  
        }  
        System.out.println("Sum = " + sum);  
    }  
}
```

Output:

Sum = 15

Program 10: Multiplication Table of a Number

```
class MultiplicationTable {  
    public static void main(String[] args) {  
        int n = 5;  
        for (int i = 1; i <= 10; i++) {  
            System.out.println(n + " x " + i + " = " + (n * i));  
        }  
    }  
}
```

```
}  
}
```

Output:

```
5 x 1 = 5  
5 x 2 = 10  
5 x 3 = 15  
5 x 4 = 20  
5 x 5 = 25  
5 x 6 = 30  
5 x 7 = 35  
5 x 8 = 40  
5 x 9 = 45  
5 x 10 = 50
```

POST LAB EXERCISE

- ✓ What is the use of if statement?

The if statement is used to check a condition.

If the condition is true, the block inside if will execute.

- ✓ Difference between if-else and else-if ladder.

- **if-else:** used to choose between **two options**.

- **else-if ladder:** used to check **multiple conditions** one by one.

- ✓ Why is switch statement used?

The switch statement is used to select one option from many choices based on a single value (like menu, day number, etc.).

- ✓ Difference between for, while, and do-while loops.

- **for loop:** used when iterations are known.

- **while loop:** checks condition first, may run 0 times.

- **do-while loop:** runs first, then checks condition.

- ✓ Which loop executes at least once?
-do-while loop executes at least once.

Result:

Thus the different control statements were executed successfully with expected output.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		